

MMCOE — Computer Engineering

Data Analytics Minor Project · 2025

CRYPDASH 2020

Crypto Market Analytics Dashboard

Full Year 2020 · 20 Cryptocurrencies · 6,708 Daily Records

Project Team

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Live Dashboard

<https://jashwantnukala.github.io/crypdash/>

1. Introduction

The global cryptocurrency market experienced one of its most volatile and transformative years in 2020. Beginning with a pre-COVID bull run, then crashing sharply in March, and ultimately closing the year with Bitcoin reaching a new all-time high, the year encapsulated the full spectrum of market behaviour — fear, recovery, and euphoria — all within twelve months.

Analyzing such large, multi-coin datasets manually is extremely difficult. Price data, volume, market capitalization, and return metrics across 20 different cryptocurrencies over 365 days produce thousands of data points that are impossible to interpret without structured analysis and visualization.

This report presents CRYPDASH 2020, a Crypto Market Analytics Dashboard that transforms raw CoinGecko historical data into clear, interactive insights. The project was built using Microsoft Excel for data preprocessing and KPI extraction, and a fully custom HTML + CSS + JavaScript web dashboard powered by Chart.js for interactive visualization and live deployment.

The dashboard helps users understand:

- Coin Performance & ROI — which cryptocurrencies grew the most in 2020
- Market Phase Behaviour — how Bull Run, COVID Crash, Recovery, and Year-End Rally shaped returns
- Risk vs. Return Analysis — volatility and standard deviation across all 20 coins
- Monthly Trends — volume, avg return, and best/worst coin each month
- Investment Signals — Strong Buy, Buy, Hold, or Avoid classification for each coin

2. Research

Cryptocurrency markets are driven by a combination of macroeconomic forces, investor sentiment, regulatory developments, and technological milestones. The year 2020 was particularly significant due to the global COVID-19 pandemic, which caused widespread financial uncertainty and paradoxically accelerated institutional interest in digital assets as an alternative store of value.

For this project, the CoinGecko historical API was used to source daily OHLCV (Open, High, Low, Close, Volume) data for 20 major cryptocurrencies throughout the full calendar year 2020. This produced 6,708 individual daily records that were then organized, cleaned, and analyzed in Microsoft Excel before being embedded into the interactive web dashboard.

2.1 Dataset Variables

The dataset was organized around the following variable categories:

- Numerical Variables: daily close price, volume, market cap, ROI, average return, standard deviation, volatility
- Categorical Variables: coin name, ticker symbol, risk level (Low / Medium / High), investment signal (Strong Buy / Buy / Hold / Avoid)
- Derived Variables: full-year ROI multiplier, avg daily return %, annual volatility %, market phase classification

2.2 Research Questions

The following research questions were formulated to guide the analysis:

- Which cryptocurrencies delivered the highest full-year ROI in 2020?
- How did different market phases (Bull Run, COVID Crash, Recovery, Year-End Rally) affect coin returns?
- What is the relationship between volatility (risk) and return across all 20 coins?
- Which months recorded the highest trading volumes and strongest avg daily returns?
- Can historical performance metrics be used to generate reliable investment signals?

3. Analysis

The analysis stage used Microsoft Excel for initial data cleaning, KPI computation, and pivot table summarization. The processed metrics were then embedded into the CRYPDASH web dashboard where Chart.js renders nine interactive visualizations. The analysis covers five key dimensions.

3.1 ROI Analysis

Full-year ROI was calculated for all 20 coins as the ratio of the closing price on 31 December 2020 to the closing price on 1 January 2020. Chainlink delivered the highest ROI at 9.98x, followed by NEM (7.17x), Cardano (6.15x), Stellar (5.17x), and CRO (4.98x). Stablecoins such as Tether and USD Coin returned near 0.07–0.08x, confirming their role as near-zero-return risk benchmarks.

3.2 Risk & Volatility Analysis

Risk was measured using two metrics: daily standard deviation of returns and average daily volatility. Coins were classified into three risk tiers — Low Risk (Bitcoin, Wrapped Bitcoin, Tether, USD Coin), Medium Risk (the majority of altcoins), and High Risk (Bitcoin SV). Investment signals were assigned based on the combination of ROI rank, volatility, and return consistency.

3.3 Market Phase Analysis

2020 was divided into four distinct market phases, each with clearly different return profiles:

Phase	Period	Avg Daily Return	Characteristic
Bull Run	Jan – Feb	+0.770%	Pre-COVID rally, strong upward momentum
COVID Crash	March	-0.546%	Market-wide collapse, avg drop of ~47%
Recovery	Apr – Aug	+0.575%	Gradual stabilisation across all coins
Year-End Rally	Sep – Dec	+0.191%	BTC hits all-time high, institutional demand

3.4 Monthly Analysis

Each of the 12 months was analyzed for average daily return, total market volume, average close price, and the best and worst performing coin. January recorded the highest avg daily return (+1.43%), while December saw the largest total volume. November 2020 was notable for XRP's spike following announcements that boosted market sentiment, making it that month's top performer.

3.5 Descriptive Statistics

Descriptive analysis was performed in Excel to compute totals, averages, max/min values, and counts across all 20 coins. Standard deviation was computed per coin to quantify return consistency, and volatility was measured as average absolute daily percentage movement. These metrics form the backbone of the risk classification and investment signal system.

4. Idea Built (Ideation)

The ideation stage focused on converting raw OHLCV data into a structured, visual, and interactive experience. The process covered data cleaning in Excel, KPI derivation, and full dashboard design and development.

4.1 Data Cleaning (Excel)

The raw CoinGecko dataset was cleaned in Microsoft Excel through the following steps:

- Removed duplicate date entries and coins with incomplete 2020 records
- Standardized all coin names and ticker symbols for consistent referencing
- Calculated derived columns: daily return %, cumulative ROI, rolling avg, std deviation, volatility
- Classified each coin's risk level and assigned investment signal based on ROI and volatility thresholds
- Verified data integrity using Excel filters and conditional formatting to highlight anomalies

4.2 Dashboard Design & Development

The dashboard was built as a single HTML file requiring no framework or build tool. All styling is done in CSS using custom properties (variables) that power a full dark and light theme system. JavaScript handles all data rendering, user interaction, and chart creation.

Key design decisions included:

- CSS Grid for all layout sections — KPI strip, card grids, phase cards, tables
- CSS Custom Properties for dual-theme switching — one attribute change on the HTML element updates the entire page
- Log-scale price range bars — necessary because coin prices span from \$0.001 (NEM) to \$19,897 (Wrapped BTC)
- Chart.js for all 9 data visualizations — horizontal bars, area lines, scatter, doughnut, and multi-line charts
- Animated counters, a scrolling ticker, and fade-up entrance animations for visual polish
- Coin detail modal — clicking any coin opens a popup with 9 stat cards and a monthly sparkline chart

5. Tools & Technologies Used

5.1 Microsoft Excel

Excel was used as the primary data preprocessing and analysis tool. Pivot tables were used to summarize total volume, average return, and coin rankings by month. Sorting and filtering helped isolate specific coins and time ranges. Conditional formatting highlighted top and bottom performers. Statistical functions (AVERAGE, STDEV, MAX, MIN) were used to compute all KPI values that appear in the dashboard.

5.2 HTML + CSS + JavaScript

The entire dashboard frontend was built using vanilla HTML5, CSS3, and JavaScript — no frameworks, no build pipeline. HTML provides the semantic structure. CSS handles theming, layout, and all animations. JavaScript manages data binding, user interactions (search, filter, sort, modal), and chart rendering.

5.3 Chart.js

Chart.js 4.4.1 was loaded via CDN and used to render all nine charts in the dashboard. Each chart is created in its own function and updates automatically when the user switches between dark and light mode.

Chart Type	Purpose
Horizontal Bar	ROI ranking for all 20 coins
Horizontal Bar	Average daily volatility per coin
Area Line Chart	Monthly average close price trend
Multi-Line Chart	Top 5 coins monthly return comparison
Bar Chart (Green/Red)	Average daily return per month
Bar Chart (Gradient)	Total market volume per month
Scatter Plot	Risk vs. Return for all 20 coins
Doughnut Chart	Risk level distribution (Low/Med/High)
Doughnut Chart	Market cap share — top 7 coins + others

5.4 Google Fonts & GitHub Pages

The dashboard uses the Syne typeface for headings and Space Mono for monospaced data labels, both loaded from Google Fonts. The completed dashboard was deployed publicly via GitHub Pages and is accessible at <https://jashwantnukala.github.io/crypdash/>

6. Key Performance Indicators (KPIs)

The following KPIs are displayed in the dashboard header strip and provide an immediate tournament-level overview of the 2020 crypto market. All values were computed from the cleaned Excel dataset.

KPI	Value
Total Daily Records	6,708
Cryptocurrencies Tracked	20
Best ROI 2020	+9.98x (Chainlink)
Worst ROI 2020	0.07x (USD Coin)
Average Daily Return	+0.39%
Total Market Phases	4
Highest Monthly Return	+1.43% (January)
Lowest Monthly Return	-0.55% (March)
Data Source	CoinGecko API 2020

7. Output & Key Findings

The final output of this project is a fully interactive web dashboard deployed on GitHub Pages. All charts, tables, and filters are functional and require no installation — the dashboard runs entirely in the browser.

7.1 Dashboard Features

- Dark / Light mode toggle with emoji indicator (🌙 / ☀️)
- Scrolling live ticker showing all 20 coins with ROI and return
- Real-time coin search bar filtering both the leaderboard and comparison table
- Recommendation filter buttons — Strong Buy, Buy, Hold, Avoid, Low Risk, High Risk
- Sortable investment leaderboard with ROI progress bars
- Top 5 performer badge cards (click to open coin detail modal)
- Coin detail modal — 9 stat cards + monthly sparkline chart for any coin
- Market phase summary cards — color-coded by phase
- Monthly breakdown table — return, volume, avg close, best/worst coin
- Price range visualization using a log scale to compare coins across all price levels
- Full 20-coin sortable comparison table with 12 metrics per coin

7.2 Key Findings

- **delivered the highest ROI of 2020 at an extraordinary 9.98x, driven by DeFi adoption and oracle demand**
- **caused the steepest single-month avg daily return of -0.55%, with most coins losing 40–60% of value**
- **saw Bitcoin break its 2017 all-time high, closing 2020 above \$29,000 and validating institutional confidence**
- **(Low Risk) offered the most predictable return profiles despite large nominal price moves**
- **(Tether, USD Coin) served as 0-volatility benchmarks — useful for risk comparison but not for growth**
- **— sustained momentum matters more than single spike events**
- **tended to have more consistent signals and performed well in the recovery phase**

8. Conclusion

This project successfully demonstrated how raw cryptocurrency market data can be transformed into meaningful, interactive insights through a structured data analytics workflow. By combining Microsoft Excel for preprocessing with a custom-built HTML/CSS/JavaScript dashboard, CRYPDASH 2020 presents the full complexity of a volatile market year in a clear and explorable format.

The study confirmed that 2020 was a year of extreme contrasts — a devastating crash followed by one of the strongest recoveries in crypto history. Coins like Chainlink, NEM, and Cardano that maintained consistent momentum across market phases delivered the best full-year returns, while high-volatility coins like Bitcoin SV underperformed relative to their risk.

This project demonstrated:

- How data analytics can simplify and illuminate complex financial datasets
- How MS Excel enables rapid KPI extraction and data cleaning without code
- How a framework-free web dashboard can deliver production-grade interactive visualizations
- How visual storytelling makes financial data accessible to any audience, technical or not

9. Future Scope

The current CRYPDASH 2020 dashboard is a strong foundation. Several extensions would significantly increase its real-world utility:

- Live API Integration — real-time CoinGecko price feeds for a live market version of the dashboard
- Python Analytics — advanced statistical modelling, correlation matrices, and backtesting using Pandas and NumPy
- Machine Learning — predictive coin performance models using historical features (volatility, momentum, volume trends)
- Portfolio Simulator — allow users to input a hypothetical investment and simulate 2020 returns coin-by-coin
- Multi-Year Dashboard — extend the dataset to cover 2018–2024 for a broader market cycle analysis
- Power BI / Tableau — enterprise-grade BI version for institutional use cases
- Cloud Deployment — integrate with a backend API for persistent user preferences and portfolio saving

10. References

1. CoinGecko Historical Data API — <https://www.coingecko.com/en/api>
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3. CoinMarketCap — 2020 Crypto Year in Review — <https://coinmarketcap.com/>
4. GitHub Pages Deployment Guide — <https://pages.github.com/>
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